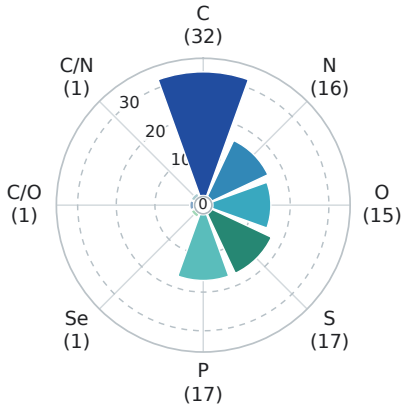
a Candidate diversity across reacting centres



b Donor form expands the search

N–H and N salts

C048 Aniline

C045 Lithium anilide

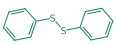
O–H and O salts

C051 Phenol

C064 Sodium phenoxide

S–H and S–S

C082 Benzenethiol



C078 Diphenyl disulfide

c Molecules retain alternative sites, pathways and endpoints

100
Candidate molecules

→

115
Site or pathway hypotheses

→

124
Proposed product options

d From recommended molecules to experimental donors

100 recommended molecules →

Expert assessment
Structural relationships · Commercial accessibility · NP / NR

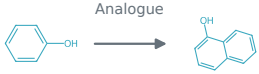
→ **100 final donors**
108 recorded input pairs

N–H model
Analogue

C048 Aniline — AP039 Indoline

Actual-pair endpoint: NP2 / NR2

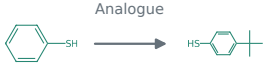
O–H model
Analogue



C051 Phenol — AP048 1-Naphthol

Actual-pair endpoint: NP2 / NR2

S–H model
Analogue



C082 Benzenethiol — AP090 4-tert-Butyl-benzenethiol

Actual-pair endpoint: NP2 / NR2

e Novelty classifications by record unit

NP Product connectivity

NR Reaction precedent